

Dice Scoring Game — Training Variant

Each game consists of two turns. In every turn, a player rolls six six-sided dice (values 1..6). Each turn is then scored against six categories; the player keeps the score from exactly one category.

Categories (per turn)

1. High-Often: the maximum value on the dice, multiplied by the number of dice showing that maximum value.
2. Summation: the plain sum of all six dice.
3. Highs-and-Lows: the product of the maximum value, the minimum value, and (maximum - minimum).
4. Only-Two: a flat bonus of 30 points if the turn contains exactly two distinct values (any two), otherwise 0.
5. All-Numbers: a flat bonus of 40 points if the turn contains every value from 1 through 6 (one of each), otherwise 0.
6. Run-of-Four: a flat bonus of 50 points if four consecutive dice (in their given order) form a run of four strictly ascending or strictly descending consecutive integers, otherwise 0.

Game Score

Within a single game, the two turns must use different categories. The game score is the maximum over all ordered pairs (i, j) with $i \neq j$ of $\text{turn1.category}[i] + \text{turn2.category}[j]$.

Data File

The data file `/root/data.xlsx` has columns: Turn (monotonically increasing), Game (each id appears twice, once per turn), R1..R6 (the six dice rolls for that turn). There are 150 games in total.

Team Partition

Team Red consists of games whose Game id is at most $N/2$ (i.e. games 1..75). Team Blue consists of games whose Game id is strictly greater than $N/2$ (games 76..150). The k -th Red game is paired against the k -th Blue game (game 1 vs game 76, game 2 vs game 77, ...).